A Lemmatized Corpus of the Pāli Canon with Critical Apparatus

Dan Zigmond
Independent Scholar, Palo Alto, CA, USA
djz@shmonk.com

Abstract

We present a lemmatized corpus of the Pāli Canon (Tipiṭaka), a 1.6 million token corpus achieving 97.7% lemmatization coverage through integration with the Digital Pāli Dictionary. Error analysis of 500 unresolved forms reveals predictable categories: 50% are potential dictionary additions, 25% are unanalyzed sandhi compounds, and 22% are metrical spelling variants. The corpus enables previously infeasible computational tasks for Pāli: training supervised morphological disambiguators, building syntactic treebanks following the Universal Dependencies framework, and conducting cross-lingual alignment with Sanskrit parallel texts. Unlike existing digital editions, this resource provides word-level lemmatization with sandhi decomposition for 42,400 compound forms, specialized handling for titles and metrical variants, and a three-witness critical apparatus recording textual variants. We compare coverage and design decisions against the Digital Corpus of Sanskrit, demonstrating comparable annotation density for a language with significantly fewer computational resources. The corpus and reproducible pipeline are released under open licenses.

Keywords: Pāli; corpus linguistics; lemmatization; morphological analysis; low-resource languages; language resources

1 Introduction

1.1 The Resource Gap for Classical Languages

Morphologically annotated corpora are foundational infrastructure for computational linguistics, yet coverage remains highly uneven across languages. While major modern languages benefit from million-token treebanks and pretrained models, classical and historical languages often lack even basic lemmatization—despite being essential for philological research, historical linguistics, and cultural heritage preservation.

Pāli exemplifies this resource gap. As a Middle Indo-Aryan language with approximately 2.7 million words of canonical text, Pāli presents a corpus comparable in size to Ancient Greek or Classical Latin. Yet while Greek and Latin have Universal Dependencies treebanks, morphological analyzers, and neural taggers, Pāli computational resources remain fragmented: digital editions exist but lack morphological annotation; dictionaries exist but lack corpus integration; morphological analyzers exist but lack corpus-scale coverage.

1.2 Computational Challenges of Pāli

Pāli presents specific challenges for NLP that distinguish it from both modern languages and better-resourced classical languages:

Rich morphology: Nouns decline for eight cases and three numbers; verbs conjugate across multiple tenses, moods, voices, and persons. A single lemma may appear in dozens of surface forms.

Pervasive sandhi: Euphonic combination at word boundaries obscures lexical units. The surface form *etadavoca* combines pronoun *etad* + verb *avoca*; *bhaborūpaṃ* joins *bhava* + *arūpaṃ*. Without decomposition, such forms cannot be lemmatized.

Orthographic variation: Regional manuscript traditions (Sri Lankan, Burmese, Thai) exhibit systematic spelling differences, complicating cross-edition alignment.

Limited training data: Unlike Sanskrit, which benefits from the Digital Corpus of Sanskrit (Hellwig, 2010) and Sanskrit Heritage Engine (Huet, 2005; Goyal and Huet, 2012), no corpus-scale annotated Pāli dataset has been available for supervised learning.

1.3 Limitations of Existing Resources

Previous Pāli computational work has been constrained by resource availability. Alfter (2017) developed a finite-state morphological analyzer achieving high precision on in-vocabulary forms, but coverage of the full canon remained incomplete due to sandhi and out-of-vocabulary items. The Digital Pāli Reader (Yuttadhammo, 2013) provides dictionary lookup with sandhi analysis but does not offer downloadable corpus-wide annotation.

In earlier work, we demonstrated that frequency-based text mining could identify textual strata in the canon (Zigmond, 2021, 2023), but concluded that "the Pāli Canon in raw form is a poor foundation for this sort of textual analysis" due to the proliferation of uninflected surface forms (Zigmond, 2023). A single lemma like *bhikkhu* (monk) appears in 270 distinct surface forms when counting declensions, compounds, and sandhi variants.

1.4 Contributions

This paper presents a lemmatized corpus that addresses these gaps. Our contributions provide new *research capabilities* for Pāli computational linguistics:

1. **Corpus-wide lemmatized analysis**: The first complete morphological annotation of the Pāli Canon, enabling vocabulary studies, distributional semantics, and text classification at scale.

2. **Reproducible evaluation benchmark**: Systematic coverage statistics and error categorization that enable comparison with other classical language resources and tracking of future improvements.

3. **Adaptable pipeline**: A modular processing architecture applicable to other Middle Indo-Aryan languages (Prakrit, Buddhist Hybrid Sanskrit) sharing similar morphological features.

4. **NLP-ready critical apparatus**: Textual variant classification that supports annotation stability analysis and cross-tradition alignment experiments.

2 Related Work

2.1 Sanskrit Computational Resources

The Digital Corpus of Sanskrit (DCS) (Hellwig, 2010) provides the closest comparison for our work. The DCS offers morphologically tagged text for over 650,000 sentences across 250+ texts, with lemmatization, POS tags, and dependency annotation. Recent extensions include the Sanskrit Sembank (Hellwig and Biagetti, 2025), which adds lexical semantic annotation integrated with Princeton WordNet synsets.

Sanskrit word segmentation and morphological tagging have benefited from neural approaches. Hellwig (2016) applied sequence models to sentence boundary detection; Krishna et al. (2018) developed datasets and models for joint segmentation and morphological parsing. These methods achieve F1 scores above 90% on standard benchmarks, demonstrating that supervised learning is viable for morphologically rich classical languages given sufficient annotated data.

Our corpus aims to provide comparable infrastructure for Pāli, enabling similar supervised approaches.

2.2 Pāli Digital Editions

Three major digital editions of the Pāli Canon exist:

Chaṭṭha Saṅgāyana (VRI/CST4): Released by the Vipassana Research Institute following the Sixth Buddhist Council (1954–1956). Widely used in computational work due to early digital availability, but lacks morphological annotation.

SuttaCentral Mahāsaṅgīti: A Sri Lankan recension with sentence-level segmentation and unique identifiers (e.g., "dn1:1.1"). The segmentation enables alignment with translations but does not include word-level annotation.

PTS/GRETIL: Digital transcriptions of Pali Text Society critical editions, manually input by the Dhammakaya Foundation (1989–1996). These remain the scholarly citation standard but lack systematic annotation.

The Digital Pāli Dictionary (DPD) (Digital Pāli Dictionary, 2026) represents a breakthrough resource: 88,350 headwords with 1,275,089 inflected form mappings and sandhi decomposition for attested compounds. Our corpus integrates the DPD to achieve corpus-wide annotation.

3 Corpus Construction

3.1 Source Integration

We collate three independent editions to establish text and record variants:

- **PTS (base text)**: GRETIL transcriptions serve as the authoritative reading, ensuring compatibility with standard scholarly citations.
- **SuttaCentral**: Provides sentence-level segmentation structure.

- **VRI**: Provides an independent textual witness from the Burmese tradition.

3.2 Preprocessing Pipeline

Source texts undergo normalization before lemmatization:

1. **Character normalization**: ṁ → ṃ; whitespace standardization
2. **Tokenization**: Regular expression matching Pāli orthography
3. **Case normalization**: Lowercase for dictionary lookup; original case preserved

3.3 Lemmatization Algorithm

For each token, we apply a cascaded lookup strategy:

1. **Direct lookup**: Query DPD inflected forms table
2. **Sandhi decomposition**: If DPD deconstructor provides analysis, recursively lemmatize components
3. **Orthographic fallbacks**: Apply normalization patterns (final -n → -ṃ, metrical lengthening)
4. **Proper noun matching**: Query Dictionary of Pāli Proper Names (Malalasekera, 1938)
5. **Mark unresolved**: Flag forms for manual review

3.4 Critical Apparatus Construction

Differences between editions are classified to support both philological analysis and annotation quality assessment:

- **Orthographic**: Predictable spelling differences (ṁ/ṃ, gemination)—normalized silently
- **Error**: SC=VRI≠PTS where PTS form absent from DPD—corrected with original preserved
- **Variant**: All forms are valid Pāli—recorded without correction

This classification enables users to assess annotation stability: forms attested across all three traditions have higher confidence than forms appearing in only one witness.

4 Evaluation

4.1 Coverage Statistics

Table 1 summarizes lemmatization results.

Table 1: Lemmatization Coverage

Metric	Value
Total tokens	1,606,492
Unique forms	127,032
Forms resolved	124,104 (97.7%)
Forms unresolved	2,983 (2.3%)
Sandhi decompositions	42,441

4.2 Error Analysis

We manually categorized 500 unresolved forms to identify systematic gaps. Table 2 shows the distribution by error category.

Table 2: Unresolved Form Categories (n=500)

Category	Count	%
Potential DPD additions	249	49.8%
Unanalyzed sandhi	126	25.2%
Metrical variants	110	22.0%
Complex compounds	7	1.4%
Proper nouns	4	0.8%
Transcription errors	4	0.8%

Potential DPD additions are forms that appear to be valid Pāli words missing from the current dictionary (e.g., rare verb forms, unusual compounds). **Unanalyzed sandhi** includes complex pronoun-verb fusions like *nibbānādhimuttohama* ("I am intent on nibbāna"). **Metrical variants** are predictable spelling changes for verse meter (vowel lengthening, consonant gemination).

Implications: Half of unresolved forms represent dictionary gaps rather than fundamental limitations. As DPD expands, coverage will improve automatically. The remaining categories are predictable and could be addressed through rule-based normalization.

4.3 Comparison with Sanskrit Resources

Table 3 compares our corpus with the Digital Corpus of Sanskrit.

The coverage gap reflects the relative maturity of Sanskrit resources and the DCS's iterative refinement over

Table 3: Comparison with DCS

	Pāli	DCS
Tokens	1.6M	4.5M
Unique lemmas	39K	90K+
Coverage	97.7%	99%+
Texts	5,764	250+
POS tags	No	Yes
Dependencies	No	Partial

15+ years. However, for the four main nikāyas (DN, MN, SN, AN), our coverage approaches the DCS benchmark, with only 95 unresolved forms. Our corpus provides a foundation for similar iterative improvement for Pāli.

4.4 Vocabulary Reduction

Lemmatization dramatically reduces vocabulary size, directly impacting downstream modeling. Table 4 shows the effect.

Table 4: Vocabulary Reduction via Lemmatization

	Surface	Lemmatized
Unique forms	127,032	39,460
Reduction ratio	–	3.22×

The 3.22× reduction means that statistical models trained on lemmatized text see over three times more examples per vocabulary item, improving generalization for rare words.

5 Applications and Reuse

The corpus enables specific computational tasks previously infeasible for Pāli:

5.1 Training Morphological Disambiguators

The lemmatized corpus provides silver-standard training data for supervised disambiguation. A concrete application: the surface form *dhammassa* is ambiguous between genitive singular ("of the teaching") and dative singular ("for the teaching"). With 15,000+ occurrences in context, the corpus provides sufficient examples to train context-aware classifiers, following approaches successful for Sanskrit (Krishna et al., 2018).

5.2 Cross-lingual Alignment with Sanskrit

Many Pāli texts have Sanskrit parallels (e.g., Pāli *Mahāparinibbāna Sutta* ↔ Sanskrit *Mahāparinirvāṇa Sūtra*). Lemmatization enables vocabulary-based alignment: the Pāli lemma *nibbāna* can be systematically mapped to Sanskrit *nirvāṇa*, supporting comparative studies of formulaic expressions and doctrinal terminology across Buddhist traditions.

5.3 Universal Dependencies Treebank Development

The lemmatized corpus provides the morphological layer required for UD annotation. Pāli's relatively fixed word order and explicit case marking make it a tractable target for treebank development. The sentence segmentation from SuttaCentral provides natural annotation units averaging 12 tokens—comparable to UD treebank sentence lengths.

6 Data Format and Availability

The corpus is released in JSON format with a Python API for programmatic access:

```
from pali import Canon

canon = Canon()

# Get lemmatized sutta
sutta = canon.get_sutta("dn1", lemmatized=True)
for segment in sutta.segments:
    for token in segment.tokens:
        print(f"{token.word} -> {token.lemma}")

# Search by lemma across corpus
results = canon.search_lemma("dhamma")
print(f"Found {results.total} occurrences")

# Generate document-term matrix
dtm = canon.document_term_matrix("dn",
                                  as_dataframe=True)
```

The API supports corpus navigation, lemma search with full-text indexing, vocabulary statistics, and export to LaTeX/PDF for critical edition typesetting.

The corpus is available at `https://github.com/dangerzig/pali-canon` under CC-BY-SA 4.0. The repository includes the complete corpus, Python package, and documentation.

7 Limitations

Lemma ambiguity: When multiple lemmas match a surface form, we select the first DPD entry. Estimated 15–20% of tokens have unrecorded alternative analyses. Context-aware disambiguation remains future work.

Partial coverage: Vinaya and Abhidhamma Piṭakas have only two-witness comparison (PTS/VRI), as SuttaCentral covers only the Sutta Piṭaka.

No POS tags: The current release includes lemmas but not part-of-speech annotation. DPD contains grammatical information that could support POS projection in future versions.

8 Conclusion

We have presented a lemmatized corpus of the Pāli Canon that brings Pāli computational resources closer to parity with better-resourced classical languages. The 97.7% overall coverage rate—and near-complete coverage for the four main nikāyas—along with systematic error analysis and integration with existing digital infrastructure, provide a foundation for supervised NLP, corpus linguistics, and philological research. By releasing the corpus and pipeline under open licenses, we aim to support iterative community improvement following the model established by the Digital Corpus of Sanskrit.

Acknowledgements

This work builds on the contributions of multiple digital Pāli initiatives. The Pali Text Society editions, digitized by the Dhammakaya Foundation and distributed through GRETIL, provided the base text. The Vipassana Research Institute (VRI) released the Chaṭṭha Saṅgāyana Tipiṭaka following the Sixth Buddhist Council, providing an independent Burmese textual witness. SuttaCentral contributed the Mahāsaṅgīti edition with sentence-level segmentation and reference IDs. The Digital Pāli Dictionary (DPD), developed by Ven. Bodhirasa, supplied the morphological database including 88,000+ headwords and 1.2 million inflected form mappings that made corpus-wide lemmatization possible. The Python library and processing pipeline were developed with coding assistance from Claude (Anthropic, 2024).

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